

MutualDTA: An Interpretable Drug–Target Affinity Prediction Model Leveraging Pretrained Models and Mutual Attention

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Cite This: *J. Chem. Inf. Model.* 2025, 65, 1211–1227



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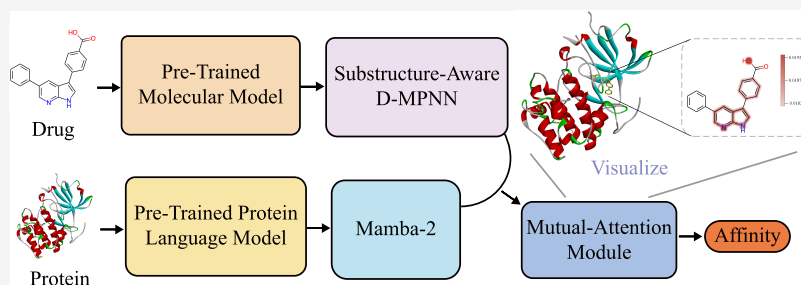
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ABSTRACT: Efficient and accurate drug–target affinity (DTA) prediction can significantly accelerate the drug development process. Recently, deep learning models have been widely applied to DTA prediction and have achieved notable success. However, existing methods often encounter several common issues: first, the data representations lack sufficient information; second, the extracted features are not comprehensive; and third, most methods lack interpretability when modeling drug–target binding. To overcome the above-mentioned problems, we propose an interpretable deep learning model called MutualDTA for predicting DTA. MutualDTA leverages the power of pretrained models to obtain accurate representations of drugs and targets. It also employs well-designed modules to extract hidden features from these representations. Furthermore, the interpretability of MutualDTA is realized by the Mutual-Attention module, which (i) establishes relationships between drugs and proteins from the perspective of intermolecular interactions between drug atoms and protein amino acid residues and (ii) allows MutualDTA to capture the binding sites based on attention scores. The test results on two benchmark data sets show that MutualDTA achieves the best performance compared to the 12 state-of-the-art models. Attention visualization experiments show that MutualDTA can capture partial interaction sites, which not only helps drug developers reduce the search space for binding sites, but also demonstrates the interpretability of MutualDTA. Finally, the trained MutualDTA is applied to screen high-affinity drug screens targeting Alzheimer’s disease (AD)-related proteins, and the screened drugs are partially present in the anti-AD drug library. These results demonstrate the reliability of MutualDTA in drug development.

INTRODUCTION

Developing drugs is time-consuming and extremely expensive. The production of a new drug, from target identification through marketing approval, can take over 12 years and cost around \$2.6 billion.¹ Unfortunately, 90% of the drugs entering clinical trials do not receive FDA approval.² Given these challenges, drug repurposing has gained significant attention in the field of drug discovery.^{3,4} By repurposing existing drugs to find potential treatments for new diseases, researchers can save substantial time and financial resources while also enhancing safety.^{5,6} A key component of drug discovery and repurposing is the identification of drug–target interactions (DTIs).⁷ Currently, DTIs are primarily characterized by the interactions between small-molecule compounds as drugs and proteins as targets.⁸ The vast chemical space to screen compounds is estimated to contain 10⁶⁰ drug-like compounds,⁹ making identifying effective drug–target pairs daunting. Facing this explosion of data on drug–target pairs, the rapid and accurate identification of novel drug–target interactions is crucial for

developing new treatments, especially for diseases like Alzheimer’s disease (AD) or COVID-19, which currently lack effective therapeutic options.

To address this challenge, the prediction of DTIs is often approached as a binary classification problem (i.e., the final result of the calculation is whether a drug and a target have an interaction),^{10–14} which neglects binding affinity values. Therefore, drug–target affinity (DTA) that reflects the strength of the interaction between drug–target pairs can provide more information than using binary categorical values to measure DTIs.¹⁵ Also, DTA is usually applied in the drug

Received: October 15, 2024

Revised: January 18, 2025

Accepted: January 20, 2025

Published: January 29, 2025



discovery pipeline to identify molecules that bind with high affinity to the target, which can be further developed into drug-like molecules (lead compounds).¹⁶ However, experimentally measuring DTA is very costly and complex, which is one of the major bottlenecks of the drug discovery process. Traditional structure-based computational methods, such as molecular docking and molecular dynamics (MD) simulations, have been widely used for affinity prediction and virtual screening.^{17,18} But they cannot be performed when the three-dimensional (3D) structures of proteins are unknown.^{4,15,19}

In recent years, thanks to the rapid increase of drug–target binding data and the decrease in computational cost, deep learning (DL) has gained remarkable achievements in various fields. DL approaches can automatically extract hidden features in the data without relying on complex feature engineering as machine learning does. Deep learning methods, which have achieved great success in the fields of computer vision (CV), natural language processing (NLP), and recommender systems, are now increasingly being used in pharmacy and chemistry.²⁰ DeepDTA²¹ was one of the first deep learning models for predicting DTA. It used convolutional neural networks (CNNs) to learn representations from the raw sequence data of proteins and drugs and employed fully connected layers in the affinity prediction block. WideDTA,²² which was an extension of DeepDTA, added two additional features to the input of the model, namely ligand maximum common structure and protein motif and structural domains. RosENet²³ used 3D structural information on drug–target complexes and then extracted complex features by 3D-CNN to predict affinity values. However, this approach is limited to known drug–target structures.

Currently, graph neural networks (GNNs) are becoming increasingly powerful and useful in the field of drug discovery and drug design.^{24–26} In the graph representing a molecule, atoms are represented by nodes and chemical bonds are represented by edges. GraphDTA²⁷ model applied GNNs and CNNs to extract features from molecular graphs and protein amino acid sequences separately. This model predicted affinity more accurately than both nondeep learning models and the comparative deep learning models. IGN²⁸ devised two independent graph convolution modules to learn intramolecular and intermolecular interactions sequentially. MGraphDTA²⁹ constructed a deep multiscale graph neural network with 27 graph convolutional layers and dense connections to capture both local and global structures of drugs. TDGraphDTA³⁰ integrated multiscale information interaction and graph optimization to predict drug–target affinity with improved accuracy and interpretability. MDCTDTA³¹ leveraged multiscale diffusion and interactive learning through CNN-Transformer blocks, Multidimensional Graph Diffusion Convolution, and a local interlayer information interaction structure, demonstrating superior performance across multiple benchmark data sets.

Besides, attention mechanisms have been used for DTA predictions. DeepAffinity+³² combined GNNs and RNNs to extract protein and drug features, integrating them with Hierarchical Attention Networks (HANs)³³ to predict DTA. MATT_DTI³⁴ constructed two attention blocks to learn correlations between atoms in a drug molecule and interaction information on drug–target pairs, while CNNs are utilized to learn the representation of drugs and targets. ML-DTI³⁵ proposed a mutual learning mechanism with multihead and position-aware attention to enhance interpretability and

generalization. ColdDTA³⁶ used graph data augmentation by removing subgraphs of drugs to improve the generalization ability and designed an attention-based feature fusion module to enhance the DTA prediction performance. MT-DTA³⁷ combined variational autoencoders and transformers to predict drug–target interactions with high accuracy and interpretability.

Moreover, large-scale pretrained protein models and pretraining tasks are increasingly employed in drug–target affinity prediction. Hu et al.³⁸ utilized graph attention networks (GATs) for a pretraining task on extensive unlabeled protein data, followed by fine-tuning the pretrained model for DTA prediction. Similarly, the GeneralizedDTA³⁹ incorporated both molecular and protein pretraining tasks and subsequently fine-tuned the pretrained model for DTA prediction. FusionDTA⁴⁰ leveraged the pretrained model ESM-1b⁴¹ to derive protein features and employed bidirectional long short-term memory (BiLSTM) layers with knowledge distillation methods to extract protein hidden features.

Although existing models have achieved promising results in affinity prediction, they still face certain limitations: (1) Recent models rely on handcrafted drug representations and one-hot encodings for protein representations, requiring models to learn from scratch. These representations are not comprehensive and lag behind the advancements in large-scale pretrained models. Despite models like GeneralizedDTA and FusionDTA incorporating pretrained tasks, the scale of pretraining is not sufficiently large. (2) Recent models overlook the role of drug substructures in affinity prediction and fail to capture the interactions between drugs and proteins from a substructural perspective. (3) Additionally, models based on CNN struggle to model the long-range dependencies between amino acids in proteins. Although transformer-based models can capture long-range dependencies, they are not efficient due to the quadratic computational overhead of the attention mechanism, which becomes a bottleneck when handling long sequences. (4) Existing affinity prediction models lack interpretability, and hard to precisely identify the binding regions between drugs and targets.

To address these issues, we propose the MutualDTA model for affinity prediction, leveraging a novel framework of “pre-trained representations + downstream feature extraction”. The key innovations of MutualDTA are as follows: First, a large-scale pretrained molecular model Uni-Mol⁴² and a protein language model ESM-2⁴³ are utilized to represent drugs and proteins comprehensively. These pretrained models have shown remarkable performance in molecular property prediction and protein 3D structure prediction but remain underutilized in the field of affinity prediction. Second, a substructure-aware drug feature extraction module is constructed to recognize drug substructures. Third, a protein feature extraction module combining a sparse autoencoder (SAE)⁴⁴ and a state-space model (SSM) Mamba^{45,46} is proposed. The sparse autoencoder improves computational efficiency by reducing intermediate dimensions, while Mamba addresses long-range dependencies and selectively captures critical amino acids in the sequence. Fourth, a Mutual-Attention Module is introduced to model the interaction between drug substructures and protein amino acids from an attention-based perspective, providing interpretability through attention scores. With these innovations, MutualDTAs “pre-trained representations + downstream feature extraction” framework is able to maximize the advantages of large-scale

pretrained models for drugs and proteins, and effectively captures drug substructures and long-term and short-term biological contextual dependencies within protein sequences.

The main contributions of this study can be summarized in the following four points. (1) We leverage a novel framework of “pre-trained representations + downstream feature extraction” for DTA prediction. Notably, we are the first to integrate both large-scale pretrained molecular and protein models for DTA prediction. (2) We construct a substructure-aware drug feature extraction module based on attention mechanism and a protein feature extraction module combining sparse autoencoder and Mamba architecture. These modules fully utilize the representations provided by pretrained models. (3) We develop the Mutual-Attention module, which enables MutualDTA to accurately identify drug–target interaction sites by highlighting specific substructures or atoms in drugs and amino acid residues in proteins. (4) We apply MutualDTA to Alzheimer’s disease-related proteins and provide medical experts with a reference to find related high-affinity drugs or lead compounds.

MATERIALS AND METHODS

Benchmark Data Sets. In this study, the data sets used as benchmarks are Davis⁴⁷ and KIBA,⁴⁸ both of which have been applied in previous studies predicting DTA. The detailed statistics of the two data sets are presented in Table 1. The

Table 1. Summary of the Davis and KIBA Data Sets

data set	drugs	proteins	number of paired samples
Davis	68	442	30,056
KIBA	2111	229	118,254

Davis data set contains more than 80% of human catalytic kinase proteins and their relevant inhibitors. It contains 30,056 dissociation constant (K_d) values calculated from interactions generated by 442 kinase proteins and 68 related inhibitors. To solve the problem of computational anomalies caused by large K_d values, Simboost²¹ converted the K_d values to logarithmic domain value (pK_d) as binding affinity values, with affinity values ranging from 5.0 to 10.8. The conversion process is shown in eq 1.

$$pK_d = -\log_{10} \frac{K_d}{10^9} \quad (1)$$

In the Davis data set, as shown in Figure 1, more than half of the affinity values (20,931 out of 30,056) are concentrated around 5. Most ligands have SMILES lengths between 50 and 90, with a maximum length of 103 and an average length of 64. The majority of protein sequences range from 500 to 1500 in length, with a maximum length of 2549 and an average length of 788.

The KIBA data set is developed from the KIBA method by integrating and optimizing the bioactivity metrics K_i , K_d and IC50 as KIBA scores. Here, the KIBA score measures the biological activity of the kinase inhibitor and is considered as the binding affinity value. SimBoost⁴⁹ filtered the KIBA data set to select only those ligands and proteins that contain at least 10 interactions, ultimately identifying 118,254 interactions between 229 proteins and 2111 ligands, resulting in a more reasonable data distribution. The KIBA scores range from 0.0 to 17.2. In the KIBA data set, as shown in Figure 1, most affinity values fall between 10 and 13. Most ligands have SMILES lengths below 100, with a maximum length of 590 and an average length of 58. Similarly, most protein sequences range from 500 to 1500 in length, with a maximum length of 4128 and an average length of 728.

Overview of MutualDTA. As illustrated in Figure 2, the proposed MutualDTA framework consists of five modules, including a pretrained molecular model, a pretrained protein language model, a substructure-aware drug feature extraction module, a protein feature extraction module, and a Mutual-Attention module. First, the Simplified Molecular Input Line Entry System (SMILES) and 3D conformation of a drug are fed into the pretrained model Uni-Mol and transformed into an undirected molecular graph. The molecular graph is then processed by the attention-based substructure-aware drug feature extraction module to derive hidden features for the drug’s substructures. Meanwhile, a protein’s raw amino acid sequence is fed into the pretrained model ESM-2, which generates protein representations at the amino acid level. These representations are then encoded by a sparse autoencoder and fed into the Mamba layer to extract higher-order features. Subsequently, the extracted features of both the drug and the protein obtained in the previous step are fed into

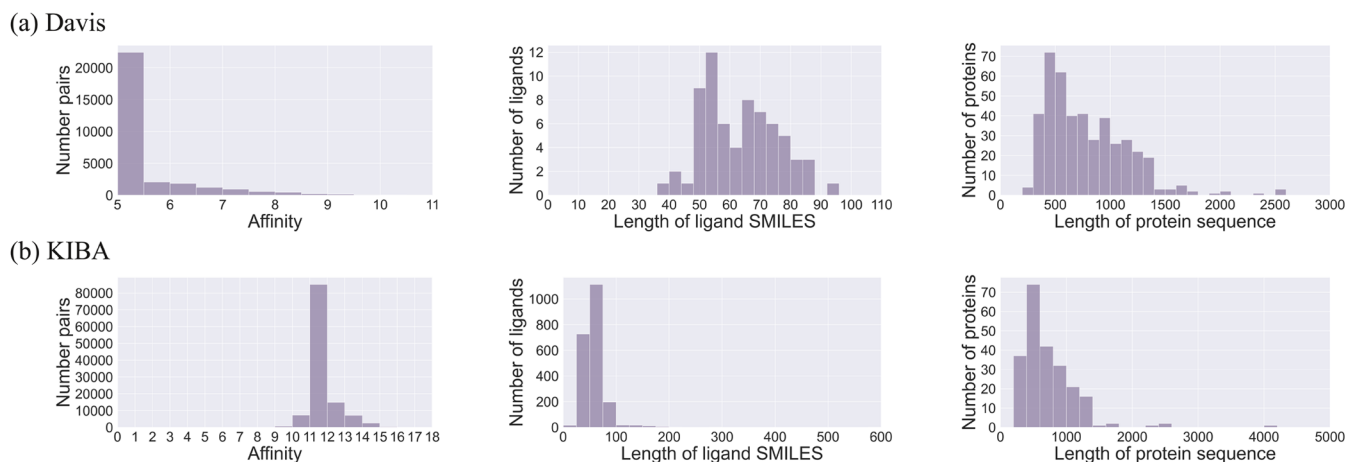


Figure 1. Distribution of binding affinity values, distribution of SMILES string lengths, and distribution of protein sequence lengths for (a) Davis and (b) KIBA data sets.

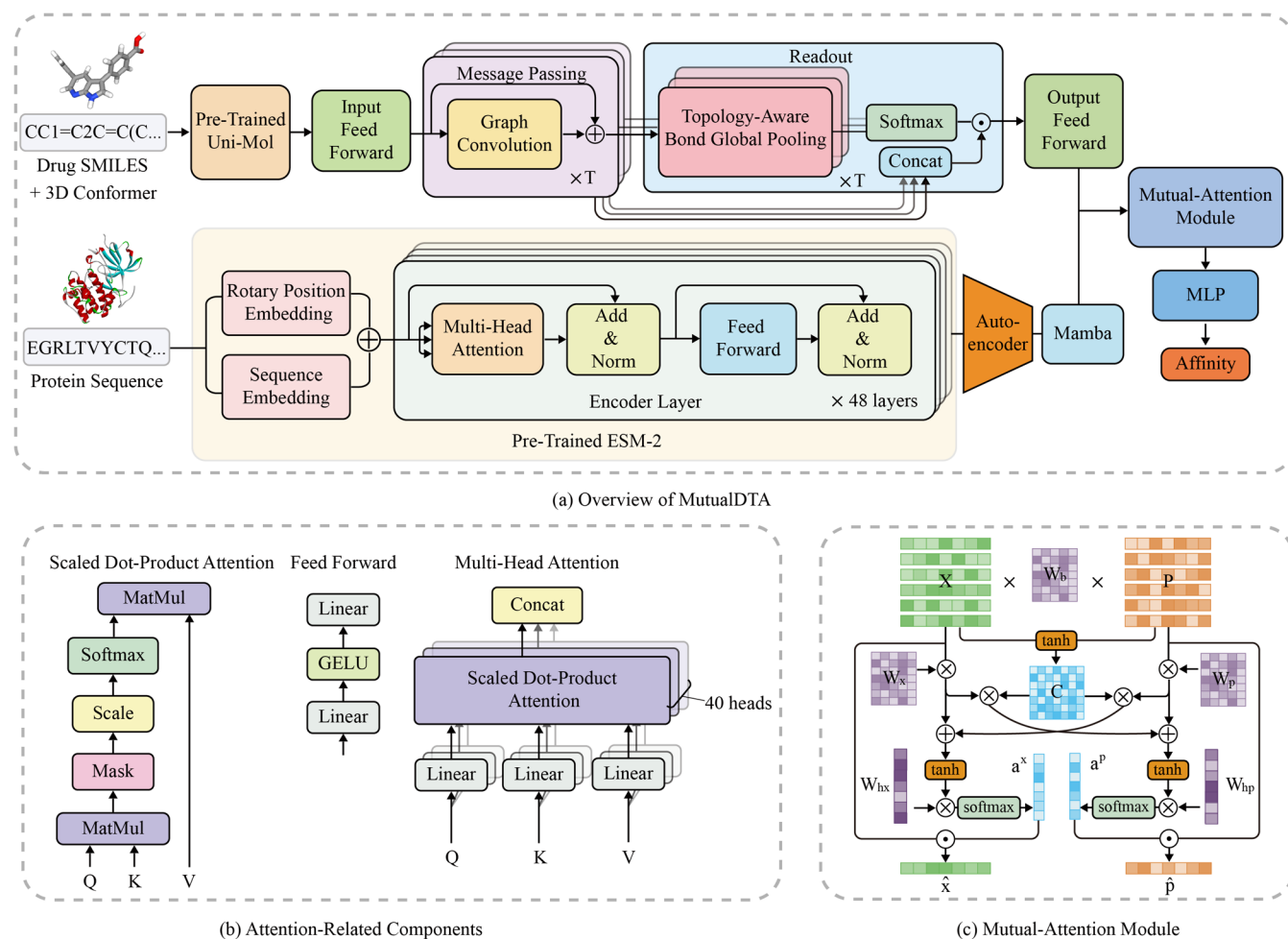


Figure 2. Illustration of the MutualDTA model proposed in this study. As depicted in (a), the MutualDTA model comprises five primary components: a pretrained molecular model, a pretrained protein language model, a substructure-aware drug feature extraction module, a protein feature extraction module combining SAE and Mamba, and a Mutual-Attention module. (b) elaborates three attention-related components: scaled dot-product attention, feed forward, and multihead attention. (c) provides a detailed depiction of the implementation details of the Mutual-Attention module.

the Mutual-Attention module, which enables the model to identify which substructures and amino acids have prominent contributions (high attention scores) in DTA. Finally, the drug hidden features and protein hidden features with attention weights are concatenated. The prediction of affinity is executed through a three-layer MLP.

Drug Representation. Currently, drugs are typically represented using one-dimensional (1D) SMILES sequences or two-dimensional (2D) topology graphs. However, those representations neglect molecular 3D structures, which are crucial for accurate drug–target affinity prediction. To incorporate drug 3D information, we introduce a powerful pretrained model, Uni-Mol, to obtain drug representations. Uni-Mol was trained on 209 million molecules and their 3D conformations, significantly enhancing the quality of molecular representations.

In our approach, SMILES sequences and 3D conformation of the molecules are fed into the Uni-Mol model to obtain atom-level representations, denoted as $X \in \mathbb{R}^{N \times 512}$, where N is the number of atoms in the molecule, and 512 indicates the dimensionality of the representations. An edge index matrix E is also constructed for each molecule to capture the interatomic adjacency relationships. This matrix defines the

source and target nodes of all edges. If atoms i and j are chemically bonded, both $[i, j]$ and $[j, i]$ are included in E . This construction process is facilitated using the RDKit toolkit.

To further enhance the molecular representations, we incorporate edge information within the molecules since the molecular representations generated by Uni-Mol do not contain edge information. We introduce six commonly used binary features to represent an edge, including indicators for single bonds, involvement in rings, and other relevant characteristics, as detailed in Table 2. These features are encoded in a vector e_{ij} representing the edge feature between atoms j and i .

Table 2. Features Characterizing the Edges

feature name	value	type	dimension
single bond	0 or 1	binary	1
double bond	0 or 1	binary	1
triple bond	0 or 1	binary	1
aromatic bond	0 or 1	binary	1
conjugated bond	0 or 1	binary	1
in ring	0 or 1	binary	1

Substructure-Aware Feature Extraction Network.

Graph Neural Networks (GNNs) are primarily used to process and learn from irregular and unstructured graph data. A molecule can be viewed as a graph consisting of atoms (nodes) and chemical bonds (edges). For a given molecule M , we construct an undirected graph $G(V, E)$, where V represents the set of atoms and E represents the set of edges. In our substructure-aware feature extraction module, we enhance a variant of the Message Passing Neural Network (MPNN) known as Directed MPNN (D-MPNN) as the backbone of our algorithm and incorporate a substructure attention mechanism to extract features from drug substructures of varying radii adaptively.

MPNN is a general framework for GNNs and is widely used in molecular representation tasks. The MPNN framework consists of two main phases: (1) the message passing phase and (2) the readout phase. The message passing phase involves k iterations, and each iteration consists of two steps: aggregation and update. The message function M aggregates the features of neighboring nodes to form a message vector $m_v^{(k+1)}$ that is passed to the target node. The update function U then combines the current node features h_v^k with the message vector $m_v^{(k+1)}$ to update the node features for the next layer.

The key distinction between D-MPNN and standard MPNN lies in the message passing mechanism. In D-MPNN, messages are associated with directed edges (bonds) rather than nodes (atoms). After the iterations, D-MPNN converts the edge-level hidden features to node-level hidden features. Before the message passing phase with D-MPNN, the hidden feature vectors of edges need to be initialized, as described in eq 2. This initialization process converts the node

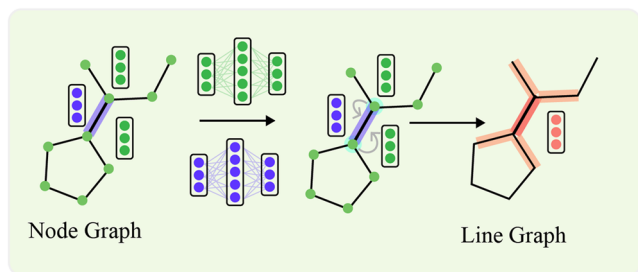


Figure 3. Input feed forward process. Converting a node graph into a line graph.

graph to the line graph, as shown in Figure 3, also referred to as Input Feed Forward in Figure 2.

$$h_{vw}^0 = \text{ReLU}(W \cdot \text{cat}(x_v, e_{vw})) \quad (2)$$

where $W \in \mathbb{R}^{h_{in} \times h_{in}}$ is a learnable weight matrix and $\text{cat}(x_v, e_{vw}) \in \mathbb{R}^{h_{in}}$ is the concatenation of the node feature vector and the edge feature vector. In the aggregation and update steps of D-MPNN, the message vector $m_{vw}^{(k+1)}$ and the edge feature $h_{vw}^{(k+1)}$ of the next layer are computed as shown in (eqs 3 and 4). In the readout phase of D-MPNN, the readout function R combines the features of the last hidden layer and obtains the representation of the entire molecular graph, as shown in eq 5.

$$m_{vw}^{(k+1)} = \sum_{j \in \{N(v) \setminus w\}} M_k(x_v, x_j, h_{jv}^{(k)}) \quad (3)$$

$$h_{vw}^{(k+1)} = U_k(h_{vw}^{(k)}, m_{vw}^{(k+1)}) \quad (4)$$

$$h_v = R(\{h_v^k | v \in G\}) \quad (5)$$

where $N(v)$ is the set of neighboring nodes of node v , M_k represents the message function, U_k represents the update function, and R is the readout function. In this study, we define the message function and update function as shown in (eqs 6 and 7).

$$M_k(x_v, x_w, h_{vw}^{(k)}) = \text{MLP}_k((1 + \epsilon^{(k)}) \cdot h_{vw}^{(k)} + x_v + x_w) \quad (6)$$

$$U_k(h_{vw}^{(k)}, m_{vw}^{(k+1)}) = \text{ReLU}(h_{vw}^0 + W_m \cdot m_{vw}^{(k+1)}) \quad (7)$$

where MLP is a multilayer perceptron and ϵ is a learnable parameter.

In the readout phase, a specialized attention mechanism is designed to align attention scores with substructures of varying radii within a drug, ultimately obtaining a substructure-level representation of the drug, as shown in Figure 4. First, for a

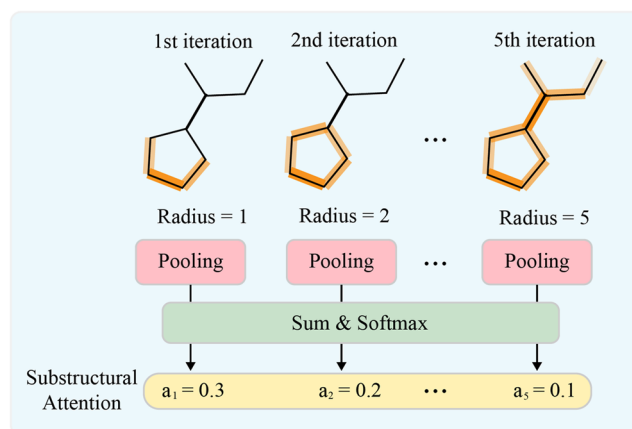


Figure 4. Readout phase.

given molecular graph, we obtain graph representation $g^{(k)}$ at the k -th iteration from the message passing phase and form a list of graph-level representations $\{g^{(0)}, g^{(1)}, \dots, g^{(k)}\}$ for graphs at different iterations. The graph representation $g^{(k)}$ is obtained through a structural-aware global pooling method known as SAGPool,⁵⁰ as shown in Figure 5, defined as (eqs 8 and 9).

$$g^{(k)} = \sum_{i=1}^n \sum_{v_j \in N(v_i)} \beta_{ji} h_{ji}^{(k)} \quad (8)$$

$$\beta_{ji} = \text{softmax}(\text{GNN}(E, X_e)) \quad (9)$$

where, E is the edge adjacency matrix, and X_e represents the hidden features of edges. The term GNN denotes the vanilla

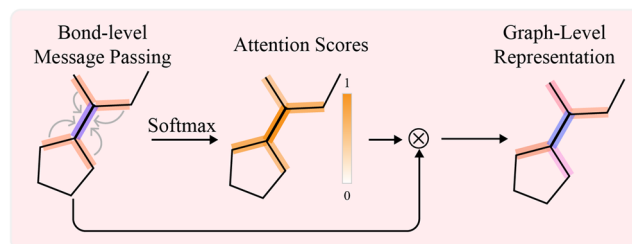


Figure 5. Topology-aware bond global pooling.

message passing operation. Using the SAGPool pooling method, we can assign a specific attention score β_{ji} to each edge feature h_{ij} , resulting in the graph-level representation $g^{(k)}$ at the k -th iteration.

Next, we calculate the attention score $\alpha^{(k)}$ for each iteration as shown in (eqs 10 and 11). The graph representation $g^{(k)}$ undergoes a nonlinear transformation to yield $g^{(k)'}$. Then, a softmax operation is applied to calculate the attention score for the graph representation at different iterations. The function $\text{Linear}(\cdot)$ denotes a linear transformation implemented as a fully connected layer. The σ function represents a nonlinear activation function. Subsequently, the attention score is assigned to the hidden edge features at different iterations to obtain the final representation of the edge hidden feature h_{ij} , as shown in eq 12.

$$g^{(k')} = \sigma(\text{Linear}(g^{(k)})) \quad (10)$$

$$\alpha^{(k)} = \frac{\exp(g^{(k)'})}{\sum_{k \in \{1, \dots, K\}} \exp(g^{(k)'})} \quad (11)$$

$$h_{vw} = \sum_{k=1}^K \alpha^{(k)} h_{vw}^{(k)} \quad (12)$$

Ultimately, we aggregate the node features by summing the edge features pointing to each node and applying nonlinear transformations. The aggregated node features contain node information from different radii and can be viewed as substructural features. Moreover, the aggregated message is combined with the original features, using a scalar β to control their balance. This is expressed as (eqs 13 and 14)

$$m_i = \sum_{v_j \in N(v_i)} h_{ji} \quad (13)$$

$$h_i = \beta m_i + (1 - \beta) x_i \quad (14)$$

where h_i represents the updated feature, m_i denotes the aggregated message, and x_i is the original feature. The scalar β adjusts the contribution of the aggregated message m_i relative to the original feature x_i .

Protein Representation and Bidirectional Transformer Encoder. Transformer⁵¹ and its variants (Bert,⁵² GPT-3⁵³) have become one of the most popular deep learning models today by leveraging a large amount of unlabeled data for self-supervised learning modeling of natural language. To exploit the advantages of the transformer-based model, we use the latest pretrained version of the bidirectional Transformer protein language model ESM-2. ESM-2 is a generalized state-of-the-art advanced protein language model. In this study, we treat protein encoding as text representation, considering protein sequences as sentences and each amino acid in the protein as a word. For a given protein, the amino acid-level representations $P \in \mathbb{R}^{L \times 5120}$ are generated by ESM-2, where L represents the total number of amino acids in the protein sequence and 5120 represents the hidden dimension of amino acids features.

Sparse Autoencoder Module. Directly utilizing the high-dimensional protein features ($L \times 5120$) for downstream DTA tasks can lead to issues such as high memory consumption, slow training speed, and convergence difficulties. To address these challenges, we employ a sparse autoencoder for protein feature extraction. Recent studies^{54,55} have demonstrated that

sparse autoencoders offer superior interpretability for large language models and yield more accurate results in downstream tasks compared to other decomposition methods.

In this study, we adopt the vanilla sparse autoencoder architecture, which consists of an encoder and a decoder. The encoder maps the input protein features $P \in \mathbb{R}^{L \times 5120}$ to a lower-dimensional hidden representation $P_{\text{hidden}} \in \mathbb{R}^{L \times 512}$ using a linear transformation followed by a ReLU activation function, shown in eq 15. The decoder reconstructs the input features from the hidden representation P_{hidden} , shown in eq 16.

$$P_{\text{hidden}} = \text{ReLU}(\text{Linear}(P)) \quad (15)$$

$$P_{\text{reconstruction}} = \text{Linear}(P_{\text{hidden}}) \quad (16)$$

where, $\text{Linear}(\cdot)$ denotes a linear layer, and ReLU is the rectified linear unit activation function.

To train the sparse autoencoder, we use a combined loss function that balances reconstruction fidelity and sparsity of the learned parameters, shown in eq 17.

$$\mathcal{L} = \|P - P_{\text{reconstruction}}\|_2^2 + \alpha \|w\|_1 \quad (17)$$

where $P \in \mathbb{R}^{L \times 5120}$ denotes the original input protein features, $P_{\text{reconstruction}} \in \mathbb{R}^{L \times 5120}$ represents the reconstructed data from the model, w is the learnable weight parameters of the Linear layers, and α is a scalar hyperparameter controlling the trade-off between the reconstruction loss and the sparsity-inducing penalty. The first term, $\|P - P_{\text{reconstruction}}\|_2^2$, is the squared Euclidean distance between the original input data and its reconstruction.

that penalizes the absolute values of the model parameters.

Mamba Layer. Mamba can be viewed as an advanced variant of the recurrent neural network (RNN), utilizing a selective state space model. While scaling linearly in sequence length, mamba is competitive with Transformer architecture in long-sequence modeling. Building upon Mamba, Mamba-2 employs a framework known as structured state space duality (SSD), achieving faster and superior performance compared to Mamba.

In this study, Mamba-2 is used to extract the features of proteins generated by the protein language model ESM-2. In real-world scenarios, the average length of amino acid sequences is relatively long. Consequently, traditional CNNs cannot capture long-sequence dependencies, and using the commonly adopted Transformer model incurs significant computational costs for such long sequences. Moreover, LSTM is less effective than Mamba in handling sequences. Therefore, we choose Mamba-2 to extract features from protein representations, enabling accurate modeling of proteins with linear complexity. As described in eq 18, for a given amino acid sequence $P = \{p_1, p_2, \dots, p_l\} \in \mathbb{R}^d$, Mamba's output is $P' = \{p'_1, p'_2, \dots, p'_l\} \in \mathbb{R}^d$.

$$P' = \text{Mamba}(P) \quad (18)$$

Mutual-Attention Module. Drug–target interactions occur at specific locations on proteins known as binding sites, typically located in pocket-like regions.¹ Identifying drugs that bind with high affinity to these sites and exhibit strong therapeutic effects is crucial for drug discovery. Various types of atomic contacts are known to constitute the physical basis of intermolecular interactions.⁵⁶ Some existing studies simply concatenate the features of drugs and proteins to model their

binding, which lacks interpretability. Thus, we construct the Mutual-Attention module based on intermolecular interactions between drug atoms and protein amino acids. It links a drug and a protein by calculating a correlation matrix for all pairs of drug atom features and protein amino acid features. Ultimately, MutualDTA is able to capture substructures or atoms and amino acid residues with high attention scores that can be treated as interaction sites. Specifically, as shown in Figure 6,

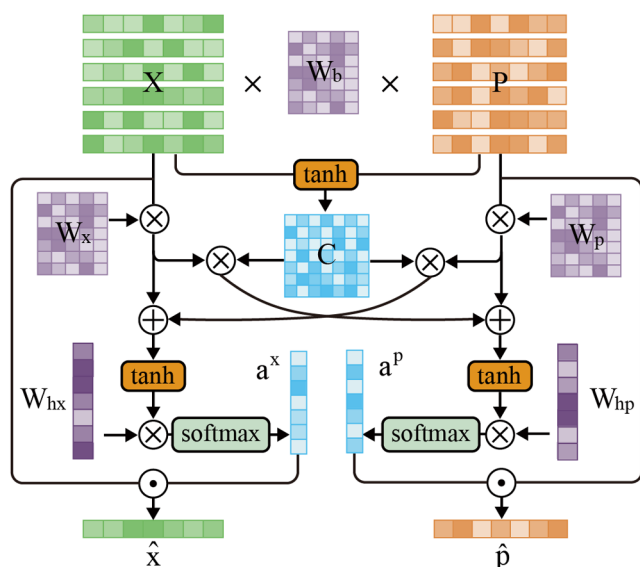


Figure 6. Illustration of Mutual-Attention. The drug hidden features X are derived from the drug feature extraction module, the protein hidden features P are derived from the protein feature extraction module, and $\hat{x}$, $\hat{p}$ are the final hidden features after incorporating the attention scores.

$X \in \mathbb{R}^{d \times N}$ is the output drug hidden features from the drug feature extraction module, $P \in \mathbb{R}^{d \times L}$ is the output protein hidden features from the protein feature extraction module, and the correlation matrix $C \in \mathbb{R}^{L \times N}$ can be calculated by the eq 19

$$C = \tanh(P^T W_b X) \quad (19)$$

where $W_b \in \mathbb{R}^{d \times d}$ contains the weights, the correlation matrix C is then considered as a feature to calculate the attention fraction of drug and protein through (eqs 20 and 21).

$$\begin{aligned} H^x &= \tanh(W_x X + (W_p P) C) \\ H^p &= \tanh(W_p P + (W_x X) C^T) \\ a^x &= \text{softmax}(w_{hx}^T H^x) \\ a^p &= \text{softmax}(w_{hp}^T H^p) \end{aligned} \quad (20)$$

where $W_x, W_p \in \mathbb{R}^{k \times d}$, $W_{hx}, W_{hp} \in \mathbb{R}^k$ are the weight parameters. $a^x \in \mathbb{R}^N$ and $a^p \in \mathbb{R}^L$ are the attention probabilities of each atom x_n in a molecule and protein amino acid p_l respectively. The correlation matrix C transforms protein attention space to drug attention space (vice versa for C^T). Based on the above attention weights, the drug and protein attention vectors are calculated as the weighted sum of the

drug hidden features and protein hidden features, as shown in eq 22

$$\begin{aligned} \hat{x} &= \sum_{n=1}^N a_n^x x_n \\ \hat{p} &= \sum_{l=1}^L a_l^p p_l \end{aligned} \quad (22)$$

Finally, as shown in Figure 6, the drug hidden features and protein hidden features ($\hat{x}$ and $\hat{p}$) are concatenated and fed into a three-layer MLP, outputting the affinity prediction values.

Evaluation Metrics. To evaluate the performance of MutualDTA, we utilize the same evaluation metrics commonly applied to benchmark state-of-the-art methods for affinity prediction, including mean square error (MSE), concordance index (CI)⁵⁷ and Regression toward the Mean (r_m^2).⁵⁸ MSE is the most popular metric for measuring the error between predicted and true results. The smaller the value of MSE, the better the regression models performance. MSE is defined as eq 23

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - y_i^*)^2 \quad (23)$$

where n is the number of drug–target pairs data samples, y_i is the true affinity value of the i -th data sample, and y_i^* is the predicted value of its affinity. CI is another typical metric to measure the probability of the concordance between the predicted and the true affinity values. The range of CI is from 0 to 1, and a larger CI means a better performance in the prediction. CI is calculated by eq 24

$$\text{CI} = \frac{1}{Z} \sum_{\delta_i > \delta_j} f(b_i - b_j) \quad (24)$$

where b_i and b_j represent the prediction score of the higher affinity δ_i and the lower affinity δ_j , respectively, Z is the normalization constant, and the step function $f(x)$ can be formulated with eq 25

$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0.5, & \text{if } x = 0 \\ 0, & \text{if } x < 0 \end{cases} \quad (25)$$

r_m^2 determines the closeness between observed and predicted values and has been extensively used by researchers to validate regression-based QSAR (Quantitative Structure–Activity Relationship) models. r_m^2 is refined and defined by two squared coefficients (r^2 and r_0^2), calculated by eq 26. Further formulation details can be found in Ref 59. An acceptable model should have an r_m^2 value greater than 0.5, and the larger the r_m^2 , the better the generalization performance of the model.

$$r_m^2 = r^2 (1 - \sqrt{r^2 - r_0^2}) \quad (26)$$

RESULTS AND DISCUSSION

This section consists of six parts. First, we compare and demonstrate the performance of MutualDTA in comparison with 12 benchmark models (KronRLS,⁶⁰ SimBoost,⁴⁹ DeepDTA,²¹ MT-DTI,⁶¹ GraphDTA,²⁷ MATT-DTI,³⁴ MRBDTA,⁶² LSTM-DTI,³⁵ ML-DTI,³⁵ FusionDTA,⁴⁰

Table 3. Data Distribution across Training Folds and Test Set

data set	Fold1	Fold2	Fold3	Fold4	Fold5	test	total
Davis	5010	5009	5009	5009	5009	5010	30,056
KIBA	19,709	19,709	19,709	19,709	19,709	19,709	118,254

MGraphDTA²⁹ and ColdDTA³⁶) on two data sets of Davis and KIBA. MutualDTA achieves the best MSE, CI, and r_m^2 on these two data sets. Second, we analyze MutualDTA's performance under cold-start settings, comparing it with 7 benchmark models. Third, ablation experiments are conducted to validate the individual contributions of the architecture modules within MutualDTA. Fourth, we provide a visual explanation for the substructure-aware drug feature extraction module, showing the module effectively identifies and distinguishes substructures within drug molecules. Fifth, as a demonstration of the Mutual-Attention mechanism's effectiveness in capturing drug–target interaction sites, a drug–target pair from the Protein Data Bank (PDB) is selected, and its 3D structure along with the high attention sites is visualized. Sixth, in an AD case study, the trained MutualDTA is applied to predict the affinity between existing drugs and 11 proteins potentially relevant to AD. The 20 drugs with the highest affinity for each protein are then selected to screen for potential AD drugs or lead compounds.

Performance Comparison with Benchmark Models.

To compare MutualDTA with benchmark models, we employ a 5-fold cross-validation methodology, consistent with previous studies.^{29,40} Specifically, the data set is divided into six parts. One part is used as an independent test set, while the remaining five parts are used for 5-fold cross-validation. The detailed sample sizes are listed in Table 3. Additionally, MutualDTA is trained and tested separately on the Davis and KIBA data sets. To ensure reliability, the final performance metrics are averaged over five independent runs. We utilize grid search hyperparameter selection to find the best configuration for MutualDTA, and the detailed results are presented in Table 4. What's more, the AdamW optimizer and

MSE, 0.908 CI, and 0.802 r_m^2 , improving by 0.001 (0.7%), 0.002 (0.2%) and 0.001 (0.1%). Overall, MutualDTA achieves optimal performance in the Davis and KIBA data sets compared to the 12 state-of-the-art affinity prediction models.

Among the 12 benchmark models, KronRLS and SimBoost are two traditional machine learning models that depend on drug and protein similarity matrices, calculated using Pubchem and Smith-Waterman algorithms. However, these two algorithms do not consider the chemical information on the atoms within a molecule and thus perform poorly compared to the other 10 deep learning models. For current deep learning methods for predicting DTA, drugs are commonly represented as SMILES character sequences or 2D topology graphs, and proteins are represented as 1D long amino acid sequences. From the comparisons in Table 5, it can be seen that most models (except LSTM-DTI and ML-DTI) using SMILES perform worse than the models using molecular graphs because SMILES sequences can not provide sufficient physicochemical and geometric structure information on drugs. For drug representation, MutualDTA takes advantage of both molecular graphs and the pretrained molecular representations, outperforming all of the benchmark models. For protein representation, models using pretrained representation, such as FusionDTA, obviously perform better than those relying on hand-crafted features or trained from scratch. Notably, MutualDTA employs the state-of-the-art pretrained model ESM-2 with 15B parameters, demonstrating superior performance compared to FusionDTA, which uses ESM-1b with 652 M parameters.

Besides, for drug feature extraction methods, models based on GNNs and attention mechanism outperform those using CNNs. CNNs cannot model potential relationships among distant atoms in a raw molecule sequence and ignore the rich global information. By utilizing substructure-aware D-MPNN based on attention mechanism, MutualDTA is able to identify the substructures of drugs and extract the features more effectively. In addition, MutualDTA applies a Mamba architecture to extract protein features. The latest Mamba-2 architecture, through its selection mechanism, is capable of retaining crucial protein information and discarding irrelevant data from the entire protein sequence, thereby ensuring the high-quality extraction of protein features. From the above comparison and analysis, it is clear that MutualDTA integrates large-scale pretrained models with well-designed feature extraction modules. Thus, it avoids various drawbacks of all benchmark models and achieves the best affinity prediction performance.

As shown in Figure 7, the left and right panels show the prediction results of MutualDTA on the test set in the Davis data set and the KIBA data set. In these two scatter plots, the *x*-axis represents the true affinity values while the *y*-axis represents the predicted affinity values. The distribution histograms at the top and right of the plots are the distribution of the predicted and true values. The diagonal lines in the plots are the regression lines of the correlation between the predicted and true values and it can be observed from Figure 7 that both regression lines are very close to $y = x$, which

Table 4. Hyperparameter Settings of MutualDTA

hyperparameters	search range	selected value
number of iterations in D-MPNN	[1, 3, 5]	3
dimension of drug hidden features	[32, 64, 128, 256]	128
dimension of target hidden features	[32, 64, 128, 256]	128
<i>d</i> in mutual-attention module	[32, 64, 128, 256]	64
<i>k</i> in mutual-attention module	[16, 32, 64]	32
learning rate	$[2 \times 10^{-3}, 2 \times 10^{-4}, 2 \times 10^{-5}]$	2×10^{-4}
dropout rate	[0.1, 0.2, 0.3]	0.3

a cosine annealing scheduler are used to adjust the learning rate during training. The MutualDTA training program is executed on a Linux CentOS 7 system with an NVIDIA GeForce RTX 3090 GPU.

As listed in Table 5, each evaluation metric of MutualDTA outperforms all of the 12 benchmark models across both data sets. MutualDTA achieves MSE, CI, and r_m^2 of 0.196, 0.908, and 0.748 on the Davis data set, with an improvement of 0.001 (0.5%), 0.003 (0.3%) and 0.005 (0.7%) over the benchmark models. In terms of the KIBA data set, MutualDTA gains 0.127

Table 5. Performance Comparison of MutualDTA and 12 Benchmark Methods on Davis and KIBA Data Sets

data set	model	drugs	proteins	MSE	CI	r_m^2
Davis	KronRLS	Pubchem Sim ^a	S-W ^a	0.379	0.871	0.407
	SimBoost	Pubchem Sim	S-W	0.282	0.872	0.644
	DeepDTA	SMILES ^b + CNN ^c	Seq ^b + CNN	0.261	0.878	0.630
	MT-DTI	SMILES + Attn ^c	Seq + CNN	0.245	0.887	0.665
	GraphDTA	Graph ^b + GNN ^c	Seq + CNN	0.229	0.893	
	MATT_DTI	SMILES + Attn and CNN	Seq + CNN	0.227	0.891	0.683
	MRBDTA	SMILES + Attn	Seq + Attn	0.216	0.901	0.716
	LSTM-DTI	SMILES + RNN ^c	Seq + CNN	0.198	0.884	0.735
	ML-DTI	SMILES + CNN	Seq + CNN	0.197	0.887	0.665
	MGraphDTA	Graph + GNN	Seq + CNN	0.207	0.900	0.710
	ColdDTA	Graph + GNN	Seq + CNN	0.250	0.884	0.652
	FusionDTA	SMILES + RNN and Attn	Pretrain ^d + RNN and Attn	0.208	0.905	0.743
	MutualDTA	Pretrain + GNN	Pretrain + SSM ^c	0.196	0.908	0.748
KIBA	KronRLS	Pubchem Sim	S-W	0.441	0.782	0.342
	SimBoost	Pubchem Sim	S-W	0.222	0.836	0.629
	DeepDTA	SMILES + CNN	Seq + CNN	0.194	0.863	0.673
	MT-DTI	SMILES + Attn	Seq + CNN	0.152	0.882	0.738
	GraphDTA	Graph + GNN	Seq + CNN	0.139	0.891	
	MATT_DTI	SMILES + Attn and CNN	Seq + CNN	0.150	0.889	0.756
	MRBDTA	SMILES + Attn	Seq + Attn	0.146	0.892	0.778
	LSTM-DTI	SMILES + RNN	Seq + CNN	0.195	0.867	0.711
	ML-DTI	SMILES + CNN	Seq + CNN	0.196	0.862	0.727
	MGraphDTA	Graph + GNN	Seq + CNN	0.128	0.902	0.801
	ColdDTA	Graph + GNN	Seq + CNN	0.189	0.864	0.670
	FusionDTA	SMILES + RNN and Attn	Pretrain + RNN and Attn	0.130	0.906	0.793
	MutualDTA	Pretrain + GNN	Pretrain + SSM	0.127	0.908	0.802

^aIndicate similarity information for drugs and proteins obtained using the Pubchem clustering server and Smith-Waterman algorithms, respectively.

^bIndicate the representation of the input data as SMILES sequences, protein amino acid sequences, and molecular graphs, respectively. ^cIndicate the methods of extracting features through CNN, self-attention mechanism, GNN, RNN and SSM, respectively. ^dIndicates obtaining representations through pretrained model.

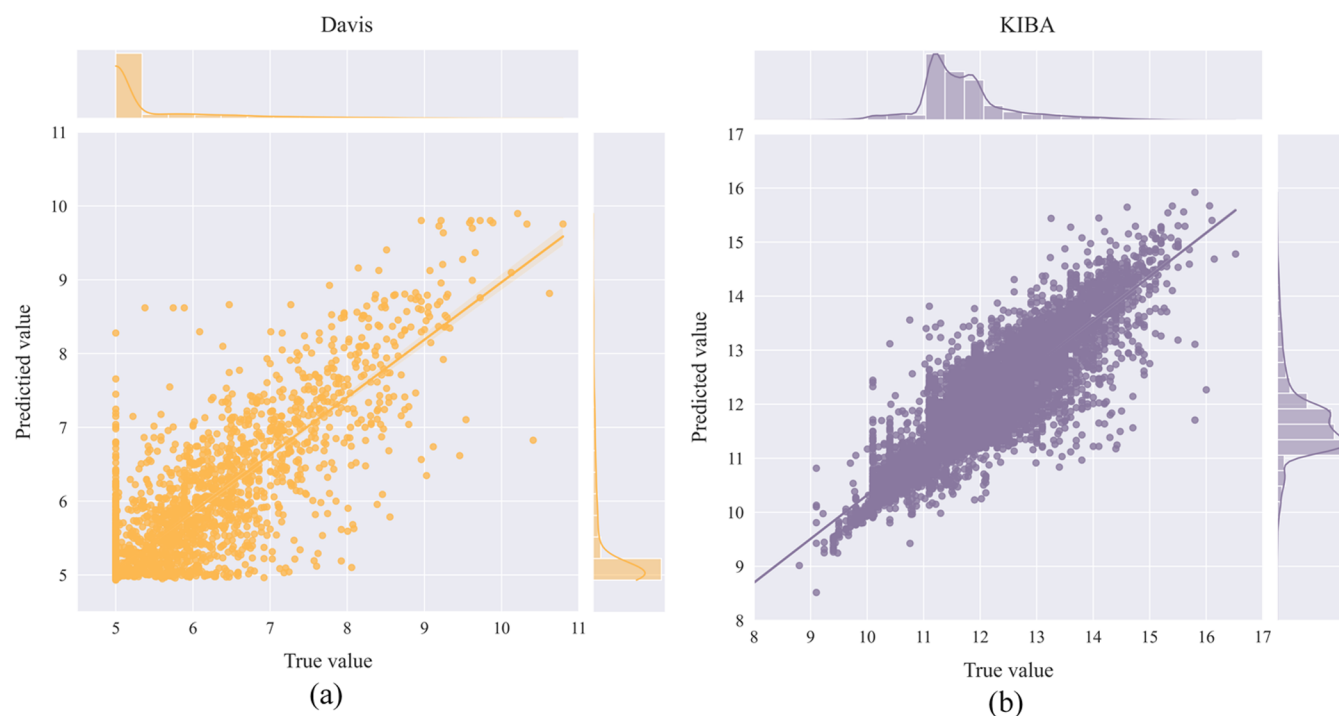


Figure 7. Binding affinities predicted by MutualDTA and the actual binding affinity values for drug–target pairs in the Davis and KIBA data sets.

indicates that the discrepancy between the predicted and true values is slight and satisfactory. The regression lines of the KIBA data set are closer to $y = x$, which means that the errors

between the predicted and actual values are smaller than the errors of the Davis data set. This phenomenon is also consistent with the results in Table 5, which shows that

Table 6. Performance Comparison Results with 7 Baseline Methods under Cold-Start Settings on KIBA Data Set

model	cold drug			cold protein			cold pair		
	MSE	CI	r_m^2	MSE	CI	r_m^2	MSE	CI	r_m^2
DeepDTA	0.824	0.743	0.388	0.395	0.773	0.484	1.073	0.570	0.039
GraphDTA	0.846	0.741	0.377	0.476	0.724	0.401	1.373	0.524	0.001
TransformerCPI	2.702	0.523	0.002	0.798	0.538	0.001	1.121	0.540	0.001
LSTM-DTI	1.010	0.706	0.273	0.407	0.760	0.466	1.153	0.512	0.002
ML-DTI	0.857	0.729	0.370	0.407	0.767	0.457	1.019	0.598	0.069
FusionDTA	0.964	0.722	0.289	0.371	0.773	0.500	0.549	0.584	0.042
MGraphDTA	0.803	0.747	0.397	0.380	0.776	0.485	1.045	0.582	0.065
MutualDTA	0.751	0.761	0.444	0.294	0.754	0.504	0.456	0.603	0.094

MutualDTA performs better on the KIBA data set. The histograms of the distributions on the top and right of the two plots show that the distributions of the predicted and true values are consistent and further demonstrate the reliability of MutualDTA.

Performance Comparison under Cold-Start Settings.

In real-world application scenarios, only a limited number of drug–target pairs are used to train the DTA prediction model. However, the number of pairs that need to be predicted in application scenarios is often extremely large, and the drugs and proteins encountered may not be seen during the training phase. To evaluate a model's generalization capability and robustness in new environments, cold-start experiments are commonly employed. In this study, we conduct three types of cold-start experiments on KIBA data set: cold drug (testing on drugs that do not appear in the training set), cold protein (testing on proteins that do not appear in the training set), and cold pair (testing on unseen drug–target pairs where neither the drugs nor the proteins are present in the training set). The data set selection criteria are detailed in the [Supporting Information](#). Our model is compared with the following 7 benchmark models: DeepDTA, GraphDTA, TransformerCPI,¹² LSTM-DTI, ML-DTI, FusionDTA and MGraphDTA.

As presented in [Table 6](#), MutualDTA achieves the best overall performance in the three cold-start experiments. Under the cold drug setting, MutualDTA gains 0.751 MSE, 0.761 CI and 0.444 r_m^2 , improving 0.052 (6.5%), 0.014 (1.9%) and 0.056 (14.4%) over the baseline models. Under the cold protein setting, MutualDTA shows advantages in the MSE and r_m^2 metrics, with increases of 0.077 (20.8%) and 0.004 (0.8%), respectively. It also achieves a comparable result of 0.754 in the CI metric. Furthermore, under the cold pair setting, MutualDTA improves by 0.093 (16.9%), 0.005 (0.8%), and 0.025 (36.2%) in MSE, CI, and r_m^2 , respectively.

We speculate that MutualDTA's superior performance in the cold-start settings is due to two key factors: (1) The incorporation of the large-scale pretrained models enables MutualDTA to provide relatively accurate 3D molecular structure representations for unseen molecules and rich property information for unseen proteins. (2) MutualDTA focuses on critical substructures thanks to the substructure-aware drug feature extraction module and the Mutual-Attention Module. The substructure-aware drug feature extraction module enables the model to learn universal substructures of drugs, while the Mutual-Attention Module facilitates the learning of general topological knowledge around the binding site, thereby enhancing its generalization capability when dealing with unfamiliar data. In conclusion, MutualDTA generalizes well and performs robustly under cold-start

settings, which proves that our model design is effective for DTA tasks in real-world application scenarios.

Ablation Experiments. Effect of Model Architecture Modules. The core innovations of our MutualDTA model are 3-fold: (1) utilizing large-scale pretrained models to generate representations for both drugs and proteins, (2) employing a substructure-aware D-MPNN based on attention mechanism and Mamba-2 architecture to extract features for drugs and proteins and (3) incorporating a Mutual-Attention module, enabling our model to capture the binding sites. To validate the effectiveness of our model's components, we design five ablation experiments.

Specifically, these experiments are (1) MutualDTA_{drug}: substituting the pretrained drug representation with drug representation applied in previous study,⁶³ (2) MutualDTA_{GCN}: replacing the substructure-aware D-MPNN with GCN, (2) MutualDTA_{protein}: substituting the pretrained protein representation with amino acid sequence one-hot embeddings, (3) MutualDTA_{BiLSTM}: substituting the Mamba layer with BiLSTM layer. (4) MutualDTA_{MLP}: replacing the Mutual-Attention module with MLP, and (5) MutualDTA_{attention}: substituting the Mutual-Attention module with self-attention. We conduct experiments on the Davis and KIBA data sets, keeping experiment settings consistent with our previous experiment of performance comparison with benchmark models. The experiment results are listed in [Table 7](#).

On both Davis and KIBA data sets, the MutualDTA model outperforms all the variant models, achieving the lowest MSE, the highest CI, and the highest r_m^2 evaluation metric. Removal of pretrained molecular representation or pretrained protein

Table 7. Results of MutualDTA Ablation Experiments on Davis and KIBA Data Sets

data set	Model	MSE	CI	r_m^2
Davis	MutualDTA _{drug}	0.279	0.874	0.585
	MutualDTA _{GCN}	0.233	0.888	0.660
	MutualDTA _{protein}	0.420	0.817	0.465
	MutualDTA _{BiLSTM}	0.222	0.894	0.687
	MutualDTA _{MLP}	0.252	0.888	0.677
	MutualDTA _{attention}	0.226	0.896	0.659
	MutualDTA	0.196	0.908	0.748
KIBA	MutualDTA _{drug}	0.215	0.844	0.643
	MutualDTA _{GCN}	0.184	0.864	0.645
	MutualDTA _{protein}	0.258	0.806	0.593
	MutualDTA _{BiLSTM}	0.185	0.854	0.702
	MutualDTA _{MLP}	0.210	0.839	0.686
	MutualDTA _{attention}	0.198	0.852	0.681
	MutualDTA	0.127	0.908	0.802

representation significantly decreased the prediction performance, indicating the pretrained models could provide high-level molecular and protein information. What's more, for the ablation study of the feature extract modules, the results show that substructure-aware D-MPNN and Mamba are more effective in extracting hidden features. Besides, MutualDTA also outperforms MutualDTA_{MLP} and MutualDTA_{attention}, which indicates that the integration of the Mutual-Attention module, significantly enhances the model's performance by capturing crucial interaction between drugs and proteins. Overall, these ablation experiments strongly support the effectiveness and rationale of each module on DTA tasks.

Effect of Sparse Autoencoder. In this section, we compare the sparse autoencoder with the PCA decomposition method and the approach without a sparse autoencoder. As shown in Figure 8, the sparse autoencoder not only accelerates the

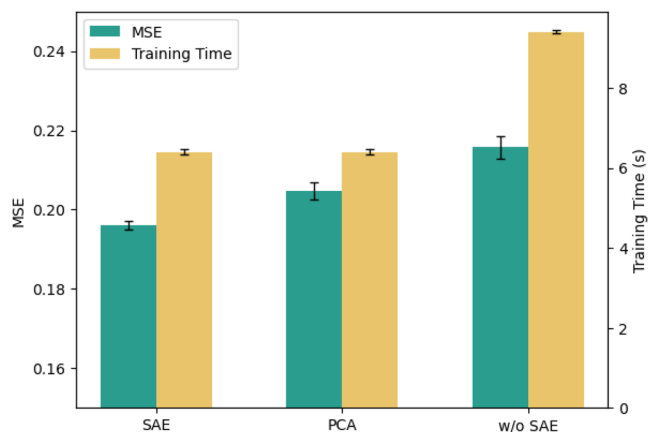


Figure 8. Comparison of MSE and average training time per epoch across decomposition methods (SAE, PCA) and approach without SAE.

training process but also improves performance by 4.2% compared to the PCA and by 9.2% compared to the approach

without the sparse autoencoder. Specifically, utilizing the SAE significantly reduces the average training time per epoch from 9.49 to 6.65 s, resulting in an approximate 30% improvement in speed. The total time required for training 5000 epochs decreases from about 13.8 h to approximately 9.25 h. These results demonstrate that the sparse autoencoder effectively utilizes features from large language models and accelerates model convergence, which is probably due to SAE reducing noise during the encoding phase.

Visual Explanation for Substructure-Aware Drug Feature Extraction Module. To verify that MutualDTA indeed identifies the substructures of drugs, we select 4 molecules that do not appear in the training set and compute the similarity matrix of their atoms by measuring the cosine similarity for the respective atom hidden vectors. These atom hidden vectors, derived from the substructure-aware drug feature extraction module (i.e., m_i in eq 13), are supposed to be the best compromise between high-level semantics and detailed spatial information.⁶³ For the atom hidden vectors m_i and m_j , the cosine similarity is calculated using eq 27.

$$\text{cosine similarity} = \frac{m_i \cdot m_j}{\|m_i\| \|m_j\|} \quad (27)$$

Figure 9 displays the four molecules alongside their atom similarity matrices and highlighted clusters of feature-similar atoms. The greater the similarity between the two atoms, the brighter the color shown in the corresponding heat map in Figure 9. As shown, different substructures in a molecule are distinguished from one another. Taking Figure 9(a) as an example, a 7-azaindole structure (atoms 6–14) and two benzene rings (atoms 0–5, atoms 15–20) are distinguished and highlighted. Besides, MutualDTA is able to distinguish the substructures with subtle structural differences, such as benzene (atoms 14–19) and pyrrolidinyl group (atoms 9–13), as shown in Figure 9(d). Moreover, MutualDTA can also recognize the same substructures in a molecule, such as the benzene rings (atoms 0–5, atoms 15–20), as shown in Figure

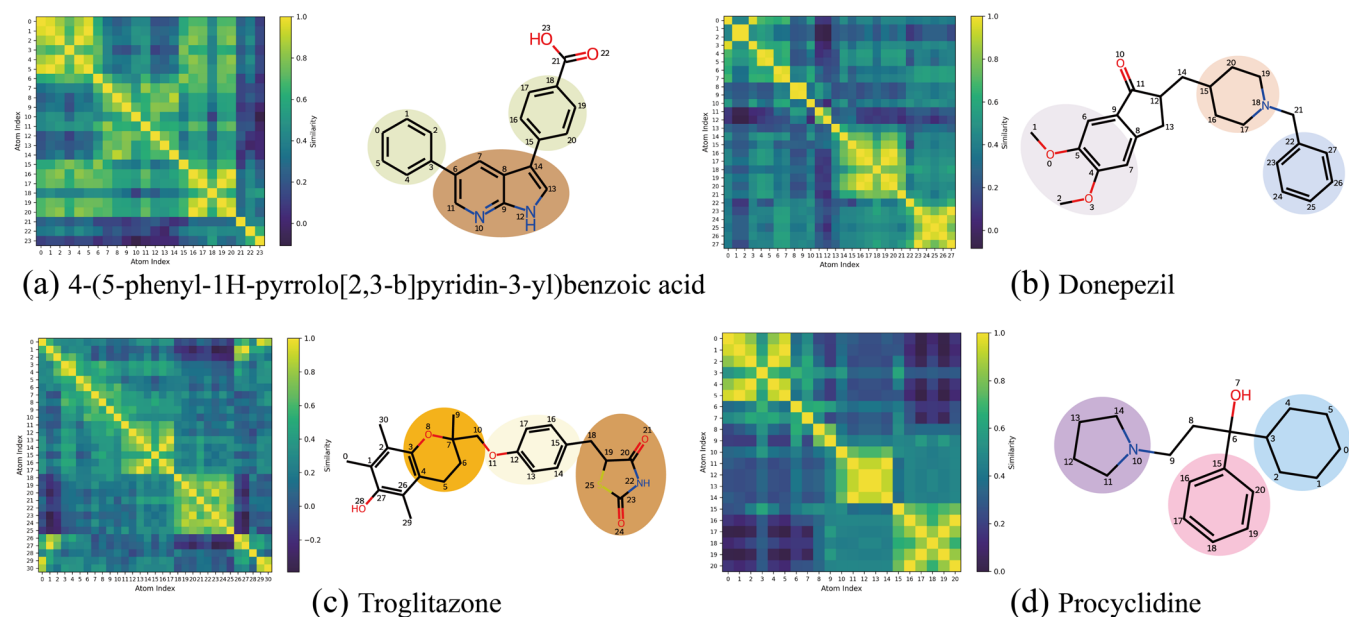


Figure 9. Heat maps of the atom similarity matrix for four molecules: (a) 4-(5-phenyl-1H-pyrrolo[2,3-b]pyridin-3-yl)benzoic acid, (b) donepezil, (c) troglitazone, and (d) procyclidine. In the molecules, clusters of feature-similar atoms are highlighted.

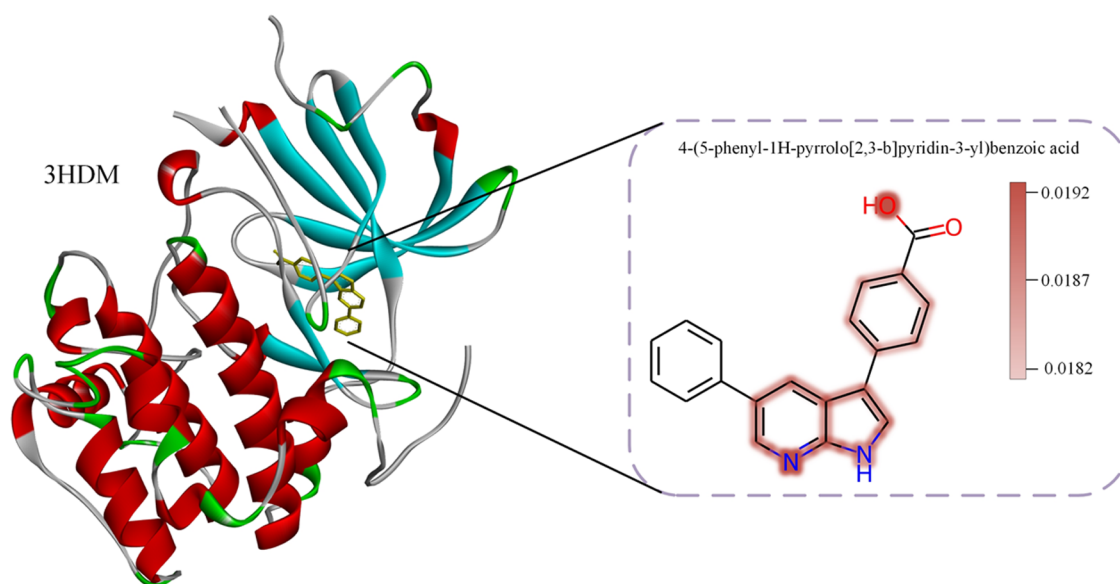


Figure 10. 4-(5-phenyl-1H-pyrrolo[2,3-b]pyridin-3-yl)benzoic acid with high attention score atoms coloring.

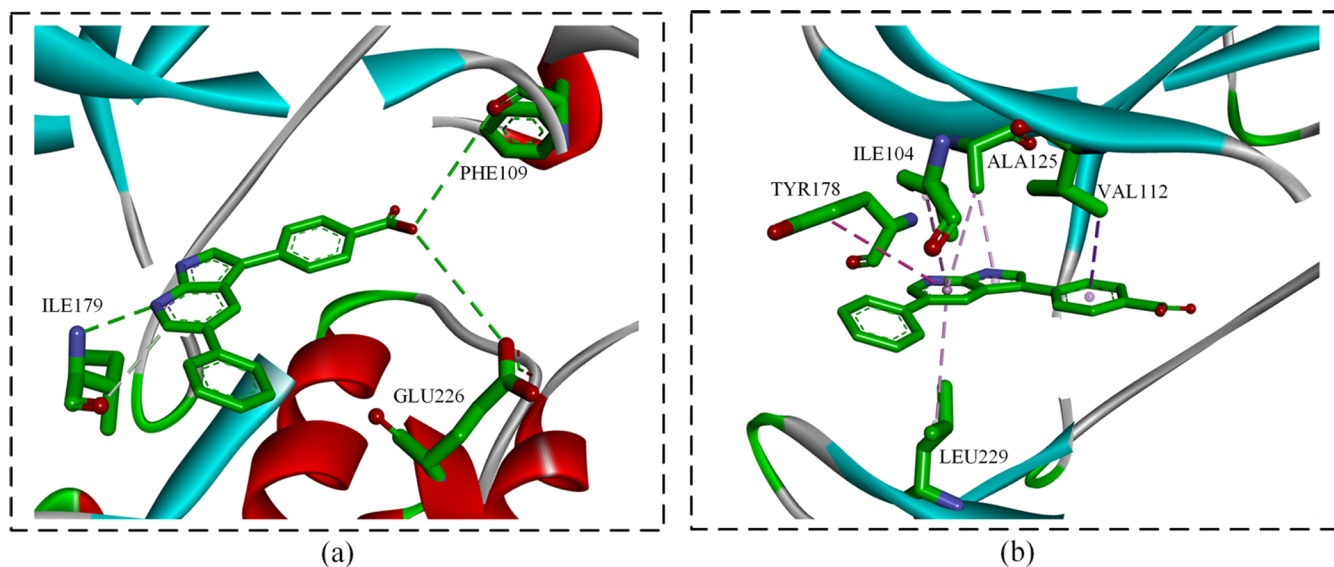


Figure 11. Actual interactions of 3HDM and 4-(5-phenyl-1H-pyrrolo[2,3-b]pyridin-3-yl)benzoic acid. (a, b) H-bond and Pi-bond interactions, respectively.

9(a). What's more, essential substructures are also included among these substructures identified by MutualDTA. For instance, highlighted substructure thiazolidinedione (atom 18–25) in Figure 9(c), plays a key role in enhancing insulin effects in Type 2 diabetes and insulin resistance and this substructure can be found in several antidiabetic drugs like rosiglitazone and pioglitazone.⁶⁴ Similarly, MutualDTA recognizes the important substructure, piperidine (atom 15–20) in Figure 9(b), with the study confirming that modifications to piperidine substructure significantly reducing the activity of Donepezil.⁶⁵

Within each highlighted substructure in Figure 9, the atom hidden vectors are similar, showing high cosine similarity in heat maps, which results in similar attention scores in the Mutual-Attention module. Conversely, atoms from different substructures exhibit dissimilar hidden vectors (indicated by darker colors in the heat maps in Figure 9), resulting in dissimilar attention scores in the Mutual-Attention module.

This ability to detect substructures aligns with the theory that the interaction between drug and target depends largely on the substructures of drug and the structure of protein sequence.^{66–69}

Visual Verification and Interpretability of the Mutual-Attention Mechanism. Accurately predicting protein–ligand binding sites is essential for drug design, aiding in the prediction of drug side effects and enhancing understanding of a protein's function.^{70–72} A binding site is considered druggable if a ligand binds with high affinity at the site and has an effective therapeutic action.⁷³ In MutualDTA, the Mutual-Attention module focuses attention on certain substructures and atoms of the molecule and amino acid residues that are crucial to the interaction between a molecule and a protein. This indicates that the crucial substructures, atoms, and amino acid residues receive high attention scores and are likely to engage in interactions. Consequently, potential binding sites are identified. A protein (PDB ID:

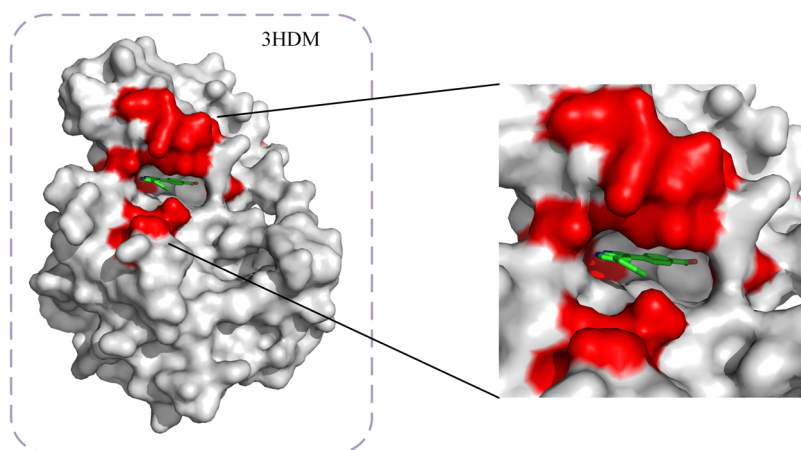


Figure 12. Binding visualization of 3HDM and 4-(5-phenyl-1H-pyrrolo[2,3-*b*]pyridin-3-yl)benzoic acid. The red part shows the amino acid region with high attention scores in 3HDM.

3HDM⁷⁴), together with its ligand 4-(5-phenyl-1H-pyrrolo[2,3-*b*]pyridin-3-yl)benzoic acid (referred to as ligand for short in the following), is selected from PDB to demonstrate MutualDTA can capture potential binding sites. 3HDM is a kinase that regulates serum and glucocorticoids, and the ligand is an inhibitor of 3HDM. First, the SMILES sequence of the ligand is converted into a molecular graph, which is then input into MutualDTA along with the amino acid sequence of 3HDM. Next, the attention weights of the ligand and 3HDM in the Mutual-Attention module are analyzed to find atoms and amino acid residues with high attention scores. If the predicted binding sites and the true binding sites overlap well, it demonstrates the effectiveness of the Mutual-Attention mechanism. As shown in Figure 10, we analyze the attention vectors of the ligand molecules in the Mutual-Attention module and mark the positions of the atoms with high attention scores in the ligand in red, darker colors indicating higher attention scores. The actual interactions between the ligand and 3HDM are shown in Figure 11. The hydrogen-bond interactions and the Pi-interactions are separated to show and analyze the interactions clearly. Accordingly, Figure 11(a) shows the H-bond interactions of the complex, while Figure 11(b) presents the Pi-interactions of the complex, including Pi-Sigma, Pi-Pi, and Pi-Alkyl interactions. Another point that needs to be noted is that the protein molecule chain in Figure 11 appears broken because when only hydrogen bonds are shown, amino acid residues with Pi-interactions are hidden, and vice versa.

First of all, it can be clearly observed in Figure 10 that the -OH group and the N atom in the six-member ring of 7-azaindole obtain the highest attention scores. In Figure 11(a), the -OH group in the ligand forms a H-bond with PHE109 and GLU226, respectively, the N atom in the six-member ring of 7-azaindole forms a H-bond with ILE179 and the C atom adjacent to this N atom also forms a C-H bond with ILE179. Similarly, in Figure 10, the two rings of 7-azaindole and benzene have high attention scores. 7-azaindole has a relatively higher attention score compared with the benzene ring. The actual interactions in Figure 11(b) show that the five-member ring of 7-azaindole forms a total of four bonds: a Pi-Pi stacked bond with TYR178, a Pi- σ bond with ILE104 bond, and a Pi-alkyl bond respectively with ALA125 and LEU229. The six-member ring of 7-azaindole forms a Pi-alkyl bond with ALA125, and the benzene forms a Pi- σ bond with VAL112. It

can be found that the high-attention score sites in the ligands captured by the Mutual-Attention module in MutualDTA overlap substantially with the actual interaction sites. This suggests that high-attention sites in the ligand are involved in intermolecular binding in the form of hydrogen-bonding or Pi-bonding interactions, both of which are essential for a stable ligand-protein affinity. Moreover, the rings with four interactions have higher attention scores and darker colors than the rings with one interaction, indicating that the Mutual-Attention module also pays attention to the information on these rings that contribute more to the interactions. Overall, the Mutual-Attention mechanism effectively recognizes the ligand positions that contribute more to the binding affinity.

Intermolecular interactions between ligands and proteins occur at specific positions in the protein, known as binding sites (also referred to as binding pockets), typically found in large and deep concavities. The attention scores in the Mutual-Attention module of all amino acid residues of 3HDM are in the range of 0.001 to 0.007, and the length of 3HDM is 373. The amino acid residues with attention scores higher than 0.004 are 101–105, 109–112, 149, 160, 162, 166, 177–179, and 182–187. As shown in Figure 12, 3HDM is displayed in surface representation by PyMol software, and these amino acid residue sites with high attention scores are colored in red. The actual binding position of the ligand to the 3HDM is shown in Figure 12, and it is evident that the colored amino acid positions are located near the binding pocket.

Alzheimer's Disease Case. Alzheimer's disease (AD) is a serious life-threatening neurodegenerative disease of the elderly that manifests clinically as a decline in language skills, memory loss, and impaired judgment. The number of patients suffering from AD will triple by 2050,^{75,76} which will not only seriously affect the quality of life of patients but also add a heavy burden to their families and society.⁷⁷ Several drugs have been proposed for the treatment of AD based on a variety of studies, including the amyloid cascade hypothesis, tau protein hyperphosphorylation, brain vascular pathology, neurotransmitter dysfunction, cell cycle dysregulation, inflammation, oxidative stress, and mitochondrial damage. Unfortunately, none of these drugs is able to cure AD.⁷⁸ Due to the difficulty and long lead time of new drug development, drug repurposing can accelerate the discovery of drugs or lead compounds for the treatment of AD. In this section, trained MutualDTA is applied to screen drugs with high affinity for AD-related

proteins from the FDA-approved drug library. It is encouraging to note that some of the screened drugs already exist in the anti-AD drug library. This result shows that the proposed MutualDTA has reliable predictive performance in drug discovery and can provide a starting point for pharmacologists to develop new AD drugs. Based on a proteome-wide association study (PWAS)⁷⁹ on AD, 11 proteins potentially relevant to AD are selected for screening high-affinity drugs from the FDA-approved drug library. The PWAS integrated AD Genome-wide association studies (GWAS) with the human brain proteome, followed by Mendelian randomization and colocalization analysis. Eleven genes are finally identified that are consistent with being causal in AD, acting via their cis-regulated brain protein abundance in the PWAS. The human protein sequences corresponding to these 11 genes in the UniProt database are listed in Table 8.

Table 8. Proteins Information Corresponding to the 11 Genes Associated with AD

no.	gene	UniProt ID	sequence length
1	CTSH	P09668	335
2	DOC2A	Q14183	400
3	ICA1L	Q8NDH6	482
4	LACTB	P83111	547
5	SNX32	Q86XE0	403
6	ACE	P12821	1306
7	RTFDC1	Q9BY42	306
8	CARHSP1	Q9Y2V2	147
9	STX6	O43752	255
10	STX4	Q12846	297
11	PLEKHA1	Q9HB21	404

FDA-approved drugs are known for their safety and reliability, making them an ideal choice for drug repurposing. In this case study, we retrieve a set of 3113 FDA-approved drugs along and an anti-AD drug library from Selleckchem Inc. (WA). Among these 3113 FDA-approved drugs, 448 are included in the anti-AD drug library, indicating that 14.4% are related to AD. The FDA-approved drugs are first converted from SMILES sequences to molecular graphs, and then combined with the amino acid sequences of 11 AD-related proteins to form drug–protein pairs, which are used as inputs for MutualDTA. The MutualDTA model applied in this study is trained on the KIBA data set. For each AD-related protein, we rank the 3,113 FDA-approved drugs according to their predicted affinity values from MutualDTA. The top 20 drugs with the highest predicted affinities are selected. Among these, 7 drugs are found in the anti-AD drug library, confirming their relevance to AD. Table 9 lists detailed information on these 7 drugs. The partial overlap between the high-affinity drugs identified by MutualDTA and the anti-AD drugs further underscores the model's reliable performance in drug discovery.

CONCLUSIONS

The application of AI in drug discovery has gained global attention. Recently, deep learning algorithms have been introduced for predicting drug–target affinity (DTA) and have shown remarkable performance. In this study, we propose a deep learning model, MutualDTA, for predicting drug–target affinity (DTA). MutualDTA outperforms 12 state-of-the-art benchmark models while offering superior interpretability. This

Table 9. 7 Drugs Screened for High Affinity and Present in the Anti-AD Drug Library

name	formula	A log P^a	HBA ^b	HBD ^c	rotatable bond ^d
Loganin	C ₁₇ H ₂₆ O ₁₀	−1.895	5	5	5
D-cycloserine	C ₃ H ₆ N ₂ O ₂	−1.788	2	2	0
L-cycloserine	C ₃ H ₆ N ₂ O ₂	−1.788	2	2	0
rutin hydrate	C ₂₇ H ₃₀ O ₁₆	−0.424	6	10	6
rutin	C ₂₇ H ₃₀ O ₁₆	−1.158	6	10	6
lithium carbonate	CHLi ₂ O ₃	−0.347	2	0	2
Ginkgolide C	C ₂₀ H ₂₄ O ₁₁	−1.753	7	3	1

^aOctanol–water partition coefficient. ^bNumber of H-bond acceptor.

^cNumber of H-bond donor. ^dNumber of rotatable donor.

success is attributed to three key factors: (1) The large-scale pretrained models accurately represent drugs and proteins. (2) The substructure-aware D-MPNN and Mamba architecture allow MutualDTA to learn the substructure-level features of a drug and the relationships between amino acids within a protein more comprehensively. (3) The constructed Mutual-Attention module models the binding of drugs and proteins from the perspective of intermolecular interactions between drug atoms and protein amino acid residues, allowing MutualDTA to be interpretable. Attention visualization analysis shows that atoms and amino acid residues with high attention scores partially overlap with the actual interaction sites. This result demonstrates the ability of the Mutual-Attention module to capture potential interaction sites and binding pocket locations, which has important implications for probing DTA and provides important support for MutualDTA in reducing the binding site search space in drug development. In addition, trained MutualDTA is applied to an AD case study to find high-affinity drugs for 11 AD-associated proteins from FDA-approved drug library, and 7 of the screened high-affinity drugs already existed in the anti-AD drug library. This means that the proposed MutualDTA can help drug developers carry out drug repurposing to treat diseases without therapeutic drugs. In general, our proposed MutualDTA for predicting drug–target affinity has excellent accuracy. It enhances interpretability by finding potential interaction sites with the help of the Mutual-Attention mechanism. What's more, the application of MutualDTA for disease drug discovery is validated.

Although our proposed method exhibits strong performance in the experimental results, there are certain limitations. Currently, MutualDTA considers only the protein 1D sequence, and incorporating protein 3D structural information could potentially enhance prediction accuracy. Moreover, computational efficiency can be further improved, for example, techniques such as knowledge distillation could be employed to reduce computational costs. In the future, this approach may be extended to compound–protein interactions and applied to drug discovery in areas such as immune diseases and cancer, presenting promising opportunities for further exploration.

ASSOCIATED CONTENT

Data Availability Statement

The data sets and resource codes are available online at <https://github.com/yydhYYDH/MutualDTA>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.4c01893>.

Overlap analysis of the Davis and KIBA data sets; cold-start data set selection criteria; external testing on the BindingDB data set; virtual drug screening targeting Alzheimer biomarkers (PDF)

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Funding

This study is supported by the National Natural Science Foundation of China (22373043), the Science and Technology Project of Gansu (21YF5GA102, 21YF5GA006, 21ZD8RA008, 22ZD6GA029), the Science and Technology Project of Lanzhou (2023332), the Science and Technology Plan Project of Chengguan District of Lanzhou (2023RCCX0005), and the Super Computing Center of Lanzhou University.

Notes

The authors declare no competing financial interest.

REFERENCES

- (1) Dhakal, A.; McKay, C.; Tanner, J. J.; Cheng, J. Artificial intelligence in the prediction of protein-ligand interactions: recent advances and future directions. *Briefings Bioinf.* **2022**, *23*, No. bbab476.
- (2) Takebe, T.; Imai, R.; Ono, S. The current status of drug discovery and development as originated in United States academia: the influence of industrial and academic collaboration on drug discovery and development. *Clin. Transl. Sci.* **2018**, *11*, 597–606.
- (3) Agamah, F. E.; Mazandu, G. K.; Hassan, R.; Bope, C. D.; Thomford, N. E.; Ghansah, A.; Chimusa, E. R. Computational/in silico methods in drug target and lead prediction. *Briefings Bioinf.* **2020**, *21*, 1663–1675.
- (4) Ezzat, A.; Wu, M.; Li, X.-L.; Kwok, C.-K. Computational prediction of drug-target interactions using chemogenomic approaches: an empirical survey. *Briefings Bioinf.* **2019**, *20*, 1337–1357.
- (5) Ashburn, T. T.; Thor, K. B. Drug repositioning: identifying and developing new uses for existing drugs. *Nat. Rev. Drug Discovery* **2004**, *3*, 673–683.
- (6) Strittmatter, S. M. Overcoming drug development bottlenecks with repurposing: old drugs learn new tricks. *Nat. Med.* **2014**, *20*, 590–591.
- (7) Zhao, Q.; Yang, M.; Cheng, Z.; Li, Y.; Wang, J. Biomedical data and deep learning computational models for predicting compound-protein relations. *IEEE/ACM Trans. Comput. Biol. Bioinf.* **2022**, *19*, 2092–2110.
- (8) Santos, R.; Ursu, O.; Gaulton, A.; Bento, A. P.; Donadi, R. S.; Bologa, C. G.; Karlsson, A.; Al-Lazikani, B.; Hersey, A.; Oprea, T. I.; Overington, J. P. A comprehensive map of molecular drug targets. *Nat. Rev. Drug Discovery* **2017**, *16*, 19–34.
- (9) Bohacek, R. S.; McMartin, C.; Guida, W. C. The art and practice of structure-based drug design: a molecular modeling perspective. *Med. Res. Rev.* **1996**, *16*, 3–50.
- (10) Zhao, Q.; Zhao, H.; Zheng, K.; Wang, J. HyperAttentionDTI: improving drug-protein interaction prediction by sequence-based deep learning with attention mechanism. *Bioinformatics* **2022**, *38*, 655–662.
- (11) Zhang, Y.; Jiang, Z.; Chen, C.; Wei, Q.; Gu, H.; Yu, B. DeepStack-DTIs: Predicting drug-target interactions using LightGBM feature selection and deep-stacked ensemble classifier. *Interdiscip. Sci.:Comput. Life Sci.* **2022**, *14*, 331–330.
- (12) Chen, L.; Tan, X.; Wang, D.; Zhong, F.; Liu, X.; Yang, T.; Luo, X.; Chen, K.; Jiang, H.; Zheng, M. TransformerCPI: improving compound-protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments. *Bioinformatics* **2020**, *36*, 4406–4414.
- (13) Lee, I.; Keum, J.; Nam, H. DeepConv-DTI: Prediction of drug-target interactions via deep learning with convolution on protein sequences. *PLoS Comput. Biol.* **2019**, *15*, No. e1007129.
- (14) Yuan, Y.; Hu, R.; Chen, S.; Zhang, X.; Liu, Z.; Zhou, G. CKG-IMC: An inductive matrix completion method enhanced by CKG and GNN for Alzheimer's disease compound-protein interactions prediction. *Comput. Biol. Med.* **2024**, *177*, No. 108612.
- (15) Bagherian, M.; Sabeti, E.; Wang, K.; Sartor, M. A.; Nikolovska-Coleska, Z.; Najarian, K. Machine learning approaches and databases for prediction of drug-target interaction: a survey paper. *Briefings Bioinf.* **2021**, *22*, 247–269.
- (16) Hughes, J. P.; Rees, S.; Kalindjian, S. B.; Philpott, K. L. Principles of early drug discovery. *Br. J. Pharmacol.* **2011**, *162*, 1239–1249.
- (17) Gilson, M. K.; Zhou, H.-X. Calculation of protein-ligand binding affinities. *Annu. Rev. Biophys. Biomol. Struct.* **2007**, *36*, 21–42.
- (18) Muegge, I.; Rarey, M. Small Molecule Docking and Scoring. In *Reviews in Computational Chemistry*; Wiley, 2001; Vol. 17, pp 1–60.
- (19) Li, H.; Gao, Z.; Kang, L.; Zhang, H.; Yang, K.; Yu, K.; Luo, X.; Zhu, W.; Chen, K.; Shen, J.; et al. TarFisDock: a web server for identifying drug targets with docking approach. *Nucleic Acids Res.* **2006**, *34*, W219–W224.
- (20) Li, Z.; Jiang, M.; Wang, S.; Zhang, S. Deep learning methods for molecular representation and property prediction. *Drug Discovery Today* **2022**, *27*, No. 103373.
- (21) Öztürk, H.; Özgür, A.; Ozkirimli, E. DeepDTA: deep drug-target binding affinity prediction. *Bioinformatics* **2018**, *34*, i821–i829.
- (22) Öztürk, H.; Ozkirimli, E.; Özgür, A. WideDTA: Prediction of Drug-Target Binding Affinity. 2019, arXiv:1902.04166. arXiv.org e-Printarchive. <https://arxiv.org/abs/1902.04166>.
- (23) Hassan-Harirou, H.; Zhang, C.; Lemmin, T. RosENet: improving binding affinity prediction by leveraging molecular mechanics energies with an ensemble of 3D convolutional neural networks. *J. Chem. Inf. Model.* **2020**, *60*, 2791–2802.
- (24) Jiang, D.; Wu, Z.; Hsieh, C.-Y.; Chen, G.; Liao, B.; Wang, Z.; Shen, C.; Cao, D.; Wu, J.; Hou, T. Could graph neural networks learn better molecular representation for drug discovery? A comparison study of descriptor-based and graph-based models. *J. Cheminf.* **2021**, *13*, No. 12.
- (25) Xiong, J.; Xiong, Z.; Chen, K.; Jiang, H.; Zheng, M. Graph neural networks for automated de novo drug design. *Drug Discovery Today* **2021**, *26*, 1382–1393.
- (26) Chuang, K. V.; Gunsalus, L. M.; Keiser, M. J. Learning molecular representations for medicinal chemistry: miniperspective. *J. Med. Chem.* **2020**, *63*, 8705–8722.

- (27) Nguyen, T.; Le, H.; Quinn, T. P.; Nguyen, T.; Le, T. D.; Venkatesh, S. GraphDTA: predicting drug-target binding affinity with graph neural networks. *Bioinformatics* **2021**, *37*, 1140–1147.
- (28) Jiang, D.; Hsieh, C.-Y.; Wu, Z.; Kang, Y.; Wang, J.; Wang, E.; Liao, B.; Shen, C.; Xu, L.; Wu, J.; et al. Interactiongraphnet: A novel and efficient deep graph representation learning framework for accurate protein-ligand interaction predictions. *J. Med. Chem.* **2021**, *64*, 18209–18232.
- (29) Yang, Z.; Zhong, W.; Zhao, L.; Chen, C. Y.-C. MGraphDTA: deep multiscale graph neural network for explainable drug-target binding affinity prediction. *Chem. Sci.* **2022**, *13*, 816–833.
- (30) Zhu, Z.; Yao, Z.; Zheng, X.; Qi, G.; Li, Y.; Mazur, N.; Gao, X.; Gong, Y.; Cong, B. Drug-target affinity prediction method based on multi-scale information interaction and graph optimization. *Comput. Biol. Med.* **2023**, *167*, No. 107621.
- (31) Zhu, Z.; Zheng, X.; Qi, G.; Gong, Y.; Li, Y.; Mazur, N.; Cong, B.; Gao, X. Drug-target binding affinity prediction model based on multi-scale diffusion and interactive learning. *Expert Syst. Appl.* **2024**, *255*, No. 124647.
- (32) Karimi, M.; Wu, D.; Wang, Z.; Shen, Y. Explainable deep relational networks for predicting compound-protein affinities and contacts. *J. Chem. Inf. Model.* **2021**, *61*, 46–66.
- (33) Yang, Z.; Yang, D.; Dyer, C.; He, X.; Smola, A.; Hovy, E. In *Hierarchical Attention Networks for Document Classification*, Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; Association for Computational Linguistics, 2016; pp 1480–1489.
- (34) Zeng, Y.; Chen, X.; Luo, Y.; Li, X.; Peng, D. Deep drug-target binding affinity prediction with multiple attention blocks. *Briefings Bioinf.* **2021**, *22*, No. bbab117.
- (35) Yang, Z.; Zhong, W.; Zhao, L.; Chen, C. Y.-C. ML-DTI: mutual learning mechanism for interpretable drug-target interaction prediction. *J. Phys. Chem. Lett.* **2021**, *12*, 4247–4261.
- (36) Fang, K.; Zhang, Y.; Du, S.; He, J. ColdDTA: utilizing data augmentation and attention-based feature fusion for drug-target binding affinity prediction. *Comput. Biol. Med.* **2023**, *164*, No. 107372.
- (37) Zhu, Z.; Yao, Z.; Qi, G.; Mazur, N.; Yang, P.; Cong, B. Associative learning mechanism for drug-target interaction prediction. *CAAI Trans. Intell. Technol.* **2023**, *8*, 1558–1577.
- (38) Hu, F.; Hu, Y.; Zhang, J.; Wang, D.; Yin, P. In *Structure Enhanced Protein-Drug Interaction Prediction using Transformer and Graph Embedding*, IEEE International Conference on Bioinformatics and Biomedicine (BIBM); IEEE, 2020; pp 1010–1014.
- (39) Lin, S.; Shi, C.; Chen, J. GeneralizedDTA: combining pre-training and multi-task learning to predict drug-target binding affinity for unknown drug discovery. *BMC Bioinf.* **2022**, *23*, No. 367.
- (40) Yuan, W.; Chen, G.; Chen, C. Y.-C. FusionDTA: attention-based feature polymerizer and knowledge distillation for drug-target binding affinity prediction. *Briefings Bioinf.* **2022**, *23*, No. bbab506.
- (41) Elnaggar, A.; Heinzinger, M.; Dallago, C.; Rehawi, G.; Wang, Y.; Jones, L.; Gibbs, T.; Feher, T.; Angerer, C.; Steinegger, M.; et al. Prottrans: Toward understanding the language of life through self-supervised learning. *IEEE Trans. Pattern Anal. Mach. Intell.* **2022**, *44*, 7112–7127.
- (42) Zhou, G.; Gao, Z.; Ding, Q.; Zheng, H.; Xu, H.; Wei, Z.; Zhang, L.; Ke, G. In *Uni-Mol: A Universal 3D Molecular Representation Learning Framework*, Eleventh International Conference on Learning Representations; ICLR, 2023.
- (43) Lin, Z.; Akin, H.; Rao, R.; Hie, B.; Zhu, Z.; Lu, W.; Smetanin, N.; Verkuil, R.; Kabeli, O.; Shmueli, Y.; dos Santos Costa, A.; Fazel-Zarandi, M.; Sercu, T.; Candido, S.; Rives, A. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **2023**, *379*, 1123–1130.
- (44) Ng, A. Sparse autoencoder. *CS294A Lecture Notes* **2011**, *72*, 1–19.
- (45) Gu, A.; Dao, T. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. 2023, arXiv:2312.00752. arXiv.org e-Printarchive. <https://arxiv.org/abs/2312.00752>.
- (46) Dao, T.; Gu, A. *Transformers are SSMs: Generalized Models and Efficient Algorithms through Structured State Space Duality* 2024, arXiv:2405.21060. arXiv.org e-Printarchive. <https://arxiv.org/abs/2405.21060>.
- (47) Lu, J.; Yang, J.; Batra, D.; Parikh, D. In *Hierarchical Question-Image co-Attention for Visual Question Answering*, Advances in Neural Information Processing Systems; NIPS, 2016.
- (48) Davis, M. I.; Hunt, J. P.; Herrgard, S.; Cicceri, P.; Wodicka, L. M.; Pallares, G.; Hocker, M.; Treiber, D. K.; Zarrinkar, P. P. Comprehensive analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2011**, *29*, 1046–1051.
- (49) He, T.; Heidemeyer, M.; Ban, F.; Cherkasov, A.; Ester, M. SimBoost: a read-across approach for predicting drug-target binding affinities using gradient boosting machines. *J. Cheminf.* **2017**, *9*, No. 24.
- (50) Lee, J.; Lee, I.; Kang, J. In *Self-Attention Graph Pooling*, International Conference on Machine Learning; PMLR, 2019; pp 3734–3743.
- (51) Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; Polosukhin, I. In *Attention is All You Need*, Advances in Neural Information Processing Systems; NIPS, 2017.
- (52) Devlin, J.; Chang, M.-W.; Lee, K.; Toutanova, K. Bert: Pre-Training of Deep Bidirectional Transformers for Language Understanding. 2018, arXiv:1810.04805. arXiv.org e-Printarchive. <https://arxiv.org/abs/1810.04805>.
- (53) Brown, T.; Mann, B.; Ryder, N.; Subbiah, M.; Kaplan, J. D.; Dhariwal, P.; Neelakantan, A.; Shyam, P.; Sastry, G.; Askell, A. et al. In *Language Models are Few-Shot Learners*, Advances in Neural Information Processing Systems; NIPS, 2020; pp 1877–1901.
- (54) Cunningham, H.; Ewart, A.; Riggs, L.; Huben, R.; Sharkey, L. Sparse Autoencoders Find Highly Interpretable Features in Language Models. 2023, arXiv:2309.08600. arXiv.org e-Printarchive. <https://arxiv.org/abs/2309.08600>.
- (55) Marks, S.; Rager, C.; Michaud, E. J.; Belinkov, Y.; Bau, D.; Mueller, A. Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models. 2024, arXiv:2403.19647. arXiv.org e-Printarchive. <https://arxiv.org/abs/2403.19647>.
- (56) Leroy, F. *Molecular Driving Forces: Statistical Thermodynamics in Biology, Chemistry, Physics, and Nanoscience*; Dill, K. A.; Bromberg, S., Eds.; Garland Science: New York, 2013; p 756 978–0-8153-4430-8.
- (57) Gönen, M.; Heller, G. Concordance probability and discriminatory power in proportional hazards regression. *Biometrika* **2005**, *92*, 965–970.
- (58) Roy, K.; Chakraborty, P.; Mitra, I.; Ojha, P. K.; Kar, S.; Das, R. N. Some case studies on application of rm2 metrics for judging quality of quantitative structure-activity relationship predictions: emphasis on scaling of response data. *J. Comput. Chem.* **2013**, *34*, 1071–1082.
- (59) Roy, P. P.; Paul, S.; Mitra, I.; Roy, K. On two novel parameters for validation of predictive QSAR models. *Molecules* **2009**, *14*, 1660–1701.
- (60) Pahikkala, T.; Airola, A.; Pietilä, S.; Shakyawar, S.; Szwajda, A.; Tang, J.; Aittokallio, T. Toward more realistic drug-target interaction predictions. *Briefings Bioinf.* **2015**, *16*, 325–337.
- (61) Shin, B.; Park, S.; Kang, K.; Ho, J. C. In *Self-Attention Based Molecule Representation for Predicting Drug-Target Interaction*, Machine Learning for Healthcare Conference; PMLR, 2019; pp 230–248.
- (62) Zhang, L.; Wang, C.-C.; Chen, X. Predicting drug-target binding affinity through molecule representation block based on multi-head attention and skip connection. *Briefings Bioinf.* **2022**, *23*, No. bbac468.
- (63) Yang, Z.; Zhong, W.; Lv, Q.; Chen, C. Y.-C. Learning size-adaptive molecular substructures for explainable drug-drug interaction prediction by substructure-aware graph neural network. *Chem. Sci.* **2022**, *13*, 8693–8703.
- (64) Day, C. Thiazolidinediones: a new class of antidiabetic drugs. *Diabetic Med.* **1999**, *16*, 179–192.

- (65) Brewster, J. T.; Dell'Acqua, S.; Thach, D. Q.; Sessler, J. L. Classics in chemical neuroscience: Donepezil. *ACS Chem. Neurosci.* **2019**, *10*, 155–167.
- (66) Wang, L.; You, Z.-H.; Li, L.-P.; Yan, X.; Zhang, W. Incorporating chemical sub-structures and protein evolutionary information for inferring drug-target interactions. *Sci. Rep.* **2020**, *10*, No. 6641.
- (67) Zhang, W.; Chen, Y.; Li, D. Drug-target interaction prediction through label propagation with linear neighborhood information. *Molecules* **2017**, *22*, No. 2056.
- (68) Zong, N.; Kim, H.; Ngo, V.; Harismendy, O. Deep mining heterogeneous networks of biomedical linked data to predict novel drug-target associations. *Bioinformatics* **2017**, *33*, 2337–2344.
- (69) Wang, L.; You, Z.-H.; Chen, X.; Li, Y.-M.; Dong, Y.-N.; Li, L.-P.; Zheng, K. LMTRDA: Using logistic model tree to predict MiRNA-disease associations by fusing multi-source information of sequences and similarities. *PLoS Comput. Biol.* **2019**, *15*, No. e1006865.
- (70) Konc, J.; Janežič, D. Binding site comparison for function prediction and pharmaceutical discovery. *Curr. Opin. Struct. Biol.* **2014**, *25*, 34–39.
- (71) Xie, L.; Xie, L.; Bourne, P. E. Structure-based systems biology for analyzing off-target binding. *Curr. Opin. Struct. Biol.* **2011**, *21*, 189–199.
- (72) Pérot, S.; Sperandio, O.; Miteva, M. A.; Camproux, A.-C.; Villoutreix, B. O. Druggable pockets and binding site centric chemical space: a paradigm shift in drug discovery. *Drug Discovery Today* **2010**, *15*, 656–667.
- (73) Hopkins, A. L.; Groom, C. R. The druggable genome. *Nat. Rev. Drug Discovery* **2002**, *1*, 727–730.
- (74) Hammond, M.; Washburn, D. G.; Hoang, T. H.; Manns, S.; Frazee, J. S.; Nakamura, H.; Patterson, J. R.; Trizna, W.; Wu, C.; Azzarano, L. M.; et al. Design and synthesis of orally bioavailable serum and glucocorticoid-regulated kinase 1 (SGK1) inhibitors. *Bioorg. Med. Chem. Lett.* **2009**, *19*, 4441–4445.
- (75) Hebert, L. E.; Weuve, J.; Scherr, P. A.; Evans, D. A. Alzheimer disease in the United States (2010–2050) estimated using the 2010 census. *Neurology* **2013**, *80*, 1778–1783.
- (76) Weiner, M. F.; Lipton, A. M. *Clinical Manual of Alzheimer Disease and other Dementias*; American Psychiatric Publishing, 2012.
- (77) Karlawish, J. How are we going to live with Alzheimer's disease? *Health Affairs* **2014**, *33*, 541–546.
- (78) Anand, R.; Gill, K. D.; Mahdi, A. A. Therapeutics of Alzheimer's disease: Past. *Neuropharmacology* **2014**, *76*, 27–50.
- (79) Wingo, A. P.; Liu, Y.; Gerasimov, E. S.; Gockley, J.; Logsdon, B. A.; Duong, D. M.; Dammer, E. B.; Robins, C.; Beach, T. G.; Reiman, E. M.; et al. Integrating human brain proteomes with genome-wide association data implicates new proteins in Alzheimer's disease pathogenesis. *Nat. Genet.* **2021**, *53*, 143–146.